need `strcasecmp()`?), then you're in good company. I've often wondered the same, and I suspect the answer is simply that this part of PHP is one of the oldest and therefore has had the most organic growth.

Well. So far, so good. We've seen functions that can be used to sort strings at a byte level, and if those strings happen to be ASCII (*not* extended ASCII!) encoded English, then we get the added bonus of those strings sorting alphabetically (which is a consequence of the design of ASCII, not any special effort on PHP's part). There is also a further refinement to the art of sorting strings in the Unicode era of programming: the application of geographically (or culturally) localized rules (rules of *collation*), which requires that we know something about the human language represented by the byte sequences in the string buffers (and more importantly, we can express that knowledge in our programs).

Collation

Collation is the comparison and organization of text, but (as I hope you are now painfully aware) strings only become human-readable text when interpreted as a certain set of characters. In order to correctly collate collections of strings as human language, we need to apply some encoding rules. As you now know, binary comparison will only correctly collate pure ASCII strings (remember the ASCII codepage was designed with collation in mind, but it was also only designed with the English language in mind); it clearly won't work for multi-byte characters. Binary comparison will also fail to correctly collate the various single-byte ISO-8859 encodings, which put the non-ASCII characters into the extended (8-bit) *0xC0–0xFF* range. For example, if we use the ISO-8859-9 Turkish language codepage, the character Ç (which lies between C and D in the Turkish alphabet) is encoded as byte *0xC7*. If sorting using a binary comparison (e.g., `strcmp()`), any words beginning with Ç would be placed after any starting with A–Z or a–z.

This isn't a problem specific to PHP; it's a problem that exists across the entire spectrum of human interaction with computer systems, it's been a problem for decades (the Encoding section of this chapter explains why), and it's the reason operating systems contain things like `libicu` (International Components for Unicode C++ library) and the `locale` utilities and libraries. For now, it's enough to know that there are system libraries for handling character sorting rules that PHP is able to hook into. Part of PHP's configuration is the `ini` setting `default_locale`, the value of which is the name of the particular system locale library required for any functions or classes that are able